

Lecture 27

$$\# \quad A = \begin{bmatrix} a & b \\ b & c \end{bmatrix}$$

Test

$$(1) \quad \lambda_1 > 0 \quad \lambda_2 > 0$$

$$(2) \quad a > 0 \quad ac - b^2 > 0$$

$$(3) \quad \text{pivots} \quad a > 0 \quad \frac{ac - b^2}{a} > 0$$

$$(4)^* \quad \underline{\underline{x^T A x > 0}}$$

$$\text{eg} \quad \begin{bmatrix} 2 & 6 \\ 6 & k \end{bmatrix} \quad \begin{array}{l} 2k - 36 > 0 \\ \underline{k > 18} \end{array}$$

$k = 18$ $\rightarrow$ positive semidefinite
 $\lambda \geq 0$

$$\# x^T A x$$

$$x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$x^T \begin{bmatrix} 2 & 6 \\ 6 & 18 \end{bmatrix} x$$

$$\rightarrow \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} 2x_1 + 6x_2 \\ 6x_1 + 18x_2 \end{bmatrix}$$

$$\Rightarrow 2x_1^2 + 6x_1x_2 + 6x_1x_2 + 18x_2^2$$

$$\Rightarrow 2x_1^2 + 18x_2^2 + 12x_1x_2$$

$$x^T A x = 2(x_1^2 + 9x_2^2 + 6x_1x_2)$$

$\#$ Now if $x^T A x$ is positive for all x_1 & x_2 then A is a positive definite matrix.

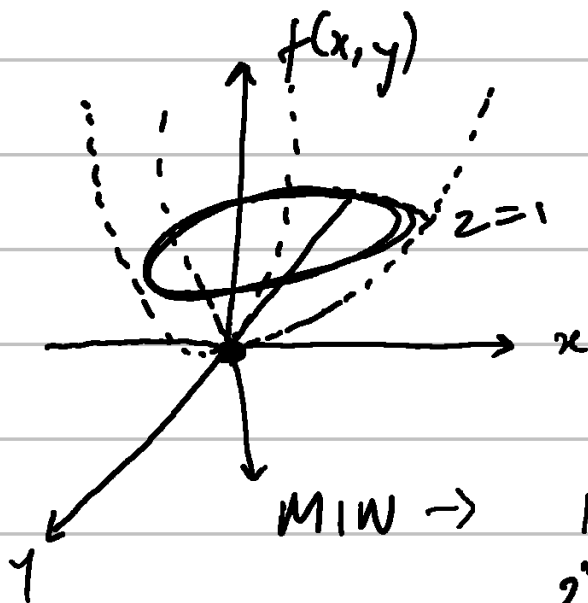
$$\# \quad A = \begin{bmatrix} 2 & 6 \\ 6 & 20 \end{bmatrix}$$

$$\vec{r} = \begin{bmatrix} x \\ y \end{bmatrix}$$

$$\vec{r}^T A \vec{r} = 2x^2 + 12xy + 20y^2$$

$$f(x, y) = 2x^2 + 12xy + 20y^2$$

$$\hookrightarrow \underbrace{2(x+3y)^2 + 2y^2}_{\text{borderline with } 18}$$



MIN $\rightarrow$ 1st derivatives are all zero
2nd derivative is positive

We get ellipse cross sections
for the definite and hyperbolas
for non definite.

NEAT

⇒ Completing the squares can be seen in elimination →

$$\begin{bmatrix} 2 & 6 \\ 6 & 20 \end{bmatrix} = \begin{bmatrix} \bar{1} & 0 \\ \bar{3} & \bar{1} \end{bmatrix} \begin{bmatrix} \textcircled{2} & 6 \\ 0 & \textcircled{2} \end{bmatrix}$$

inside factors square multipliers

$$f(x, y) = 2 \cdot [y]^2 + 2 \cdot [x + 3y]^2$$

Hence, positive pivots give us all the test for positive definiteness of a matrix.

Positive definiteness $\rightarrow$ Matrix of 2nd derivatives is ve definite.

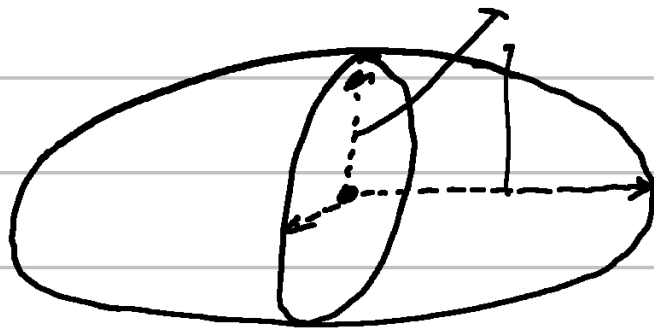
$$\begin{bmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{bmatrix}$$

3x3 example

$$A = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix} \quad \text{yes for} \\ \text{ve definite}$$

$$\begin{aligned} x^T A x &= 2x_1^2 + 2x_2^2 + 2x_3^2 - 2x_1x_2 - 2x_2x_3 \\ &= \underbrace{(x_1 - x_2)^2} + \underbrace{x_1^2} + \underbrace{(x_2 - x_3)^2} \end{aligned}$$

In higher dimensions, such as in the previous case, if we cut through a cross section, we get an ellipsoid.



- The different dimensions of the ellipsoid correspond to different eigenvalues & eigenvectors.
- All same means that the ellipsoid would be circle, sphere.